

Playing the Spread: Worker Reliability and Organizational Performance

Abstract

Workers have good days and bad days, but strategic human capital management research overwhelmingly focuses on mean performance. This paper examines whether organizations recognize and price worker volatility when they can measure it, and whether they should. Using data from professional basketball, I document both wage and production penalties for volatile workers. After separating persistent volatility from playing-time noise, more volatile workers earn less on the margin where contracts are negotiable. A difference-in-differences design then exploits the quasi-random timing of absences as an identifying shock. Holding mean performance fixed, losing a reliable worker inflicts a larger penalty on organizational outcomes than losing a volatile one. I show that the steadiest workers are eighty percent as costly to lose as a top-quintile player (near-stars through superstars), at forty percent of the salary. Though the production benefits of reliability are contemporaneous, the wage premium exists only for workers with a track record of consistency. If organizations learn a worker's consistency before the market does, this gap may represent an opportunity for competitive advantage.

Keywords: strategic human capital; performance management; compensation; labor markets; strategic factor markets

1. Introduction

On June 11, 2013, the Miami Heat lost in Game 3 of the National Basketball Association (NBA) finals by 36 points. LeBron James – the league’s most valuable player, in the midst of perhaps the greatest individual season in basketball history, leading a super-team with the NBA’s best record – scored 15 points on 21 shot attempts. It was his worst performance of the season. Advanced statistics suggest he played at a level undeserving of a roster slot. With the benefit of hindsight, we know that this was an aberration: the Heat would go on to win the best-of-seven championship series in Game 7 behind 37 points from their superstar. But, in all likelihood, there was at least one person among Game 3’s 14 million television viewers who had never watched professional basketball before. A neophyte, lacking context, might reasonably have concluded that one of the greatest players of all time was a mediocre talent on an over-matched team.

Though absurd in the context of professional basketball, the measurement and management of stochastic performance is a material and under-addressed challenge. Organizations routinely evaluate workers on singular or sparse events, despite evidence that these are noisy signals with limited predictive value. Hiring, retention, termination, promotion, and performance bonus decisions respond to periods of recent over-performance and punish underachievers. And yet, bad days are all too common. Whether because the baby won’t sleep, the weather won’t cooperate, or the muse won’t sing, individual performance is subject to nadirs at varying scales and intervals. Meta-analyses suggest that roughly half of variation in performance outcomes in both lab and field settings is attributable to within-person volatility, rather than between-person differences (Dalal et al. 2020; Podsakoff et al. 2019).¹

However, research on organizations almost exclusively focuses on the individual worker’s mean or the dispersion of worker means across a workforce or organization. This omission is particularly important for strategic human capital management. Unlike other assets, human capital

¹In the case of professional basketball, 76–92% of total game-over-game variation is within player-season. Specifically, this is variation in performance among NBA starters (639 players, 143,064 player-games, 2011–2023) for rate-adjusted Game Score (76%), BPM (86%), or plus/minus (92%).

is embodied in workers who firms do not own, who can leave of their own volition, and who can bargain over the value they create (Coff 1997, 1999). Decision-makers must identify which worker features create value for the organization and determine how those characteristics are valued on the labor market. Where there is a gap in the private and public valuations of a worker, the organization can extract value and build a sustainable competitive advantage. Traditionally, this is the result of bundles of HR practices or the development of firm-specific human capital (Becker & Gerhart 1996; Bidwell 2011; Chadwick & Dabu 2009). If the spread of a worker's performance has value distinct from the mean, it would represent an additional (and novel) wedge between labor-market price and organizational value.

This paper examines whether there is a labor market premium for reliable workers, whether worker volatility matters for organizational performance, and whether there exists a gap between the two from which a strategic and well-informed organization might extract an advantage. I answer these questions using data from the National Basketball Association, one of the few research settings in which competition occurs under a set of common rules and objectives and performance data is standardized, plentiful, and public.

I first show that organizations price consistency among workers whose compensation is negotiated rather than fixed by the league's minimum and maximum salary rules. There is also some evidence for sequential evaluation: organizations first pay for mean performance, then bargain over reliability at the margins. I then exploit the quasi-random timing of worker absences to estimate the corresponding production value of reliability. Holding mean performance fixed, losing a more reliable worker imposes a larger penalty on team outcomes (point differential and win probability) than losing a more volatile one. The implied wage and production tradeoffs are consistent in direction and magnitude, consistent with pricing which reflects the organizational value of reliability.

Though the production benefits of reliability are contemporaneous, the wage premium exists only for workers with a track record of consistency. In environments where organizations could plausibly understand a worker's consistency before the market does, I argue that this gap represents

an opportunity for competitive advantage. Further, the top quintile of reliable workers produce organizational value commensurate with the value generated by superstars at a fraction of the wages. In a setting with substantial resource constraints, this surplus can be reinvested elsewhere for even greater returns, so organizations that excel at identifying reliable workers should be able to consistently outperform those that do not.

1.1. Whither Stochastic Performance?

Since 2000, six top management journals² have published 316 empirical papers containing any mention of human resources or human capital management alongside any mention of performance or productivity³. This corpus contains a wealth of measures and methods, but I do not find a single paper using individual, within-person, intertemporal variance as a focal outcome or treatment variable. This is not to say that the role of volatility is unknown. An adjacent tradition in industrial and organizational psychology treats performance variability as a distinct dimension of worker performance (Dalal et al. 2020; Sackett et al. 1988) and finds that it interacts with compensation (Awtrey et al. 2021; Barnes & Morgeson 2007). Some authors acknowledge the absence of volatility in their analysis (De Figueiredo et al. 2013). Others include it in their theoretical specification but not their empirical findings (Aral et al. 2012; Battiston et al. 2025; DeVaro 2006).⁴ It is possible that this is a costless omission. However, stochastic performance can impose a non-negligible (Lazear 2004) and sometimes catastrophic (Perrow 1984) cost on the organization. The extent to which these costs impact organizational performance and, therefore, strategy is largely unexplored.

Recent formal work sets out scope conditions for worker volatility and strategy. Tucker (2026) represents worker performance as a distribution, rather than a scalar, and shows that the organization's preference for volatility depends on scale, resources, workforce composition, and

²AMJ, AMR, ASQ, Management Science, Organization Science, SMJ; chosen because these are the most common outlets for papers on organizations and strategy

³On Web of Science, the query is (human resources OR human capital) AND (productivity OR performance); Review conducted in Spring of 2026

⁴Only a handful of researchers concerned with the effectiveness of service operations have assessed the effect of policies, interventions, and experience on worker volatility (Bavafa & Jónasson 2021, 2024; Lu et al. 2022). However, because this work is addressed to the operations management literature, it does not show up in the initial literature search and may not be readily apparent to HR researchers or practitioners.

internal organization. Two organizations can value the same mean performance very differently, depending on the volatility of the worker. Two workers with the same mean can generate very different value, depending on how the tails of their performance distribution interact with the organization. At the limit, the best worker in the world might be the wrong person for the job if the organization cannot survive long enough to take advantage of their sporadic brilliance. This approach implies two distinct quantities: an organization-specific valuation for a worker's performance (both mean and distribution) and a labor market valuation for the worker's observable performance (plus the effects of bargaining, signaling, and other pressures). I estimate and compare both.

In addition to providing empirical support for the idea that worker volatility is strategic, this also offers insight into the somewhat startling absence of worker volatility from research on strategy and organizations. The gap is consistent with at least three candidate explanations. Perhaps worker volatility simply doesn't matter to organizational outcomes. It might be priced into wages already. Structures and routines might provide buffering and insulation (Orton & Weick 1990; Weick 1976). Or the law of large numbers might provide most organizations with statistical insurance against extreme draws. If this is the case, one would expect to find evidence that worker volatility has a null effect on organizational performance or is accurately priced across organizations.

We might instead take the lack of attention paid to worker volatility as evidence of a managerial (and academic) blind spot. After all, individuals are known to struggle to separate signal from noise or draw the correct conclusion from stochastic signals (Aldrich & Ruef 2018; Benson et al. 2019; Denrell 2005; Denrell et al. 2023; Lagziel & Lehrer 2019; Lazear 2004; Liu & Tsay 2021; March & Sutton 1997).⁵ This would suggest no wage premium and a material organizational benefit to employing reliable workers.

Finally, an organization-specific reading of Morgan's Canon⁶ suggests that we not attribute

⁵The best-documented case is promotions and the "Peter Principle," which observes that newly promoted employees are likely to underachieve at some point because they were promoted on the basis of a period of unusually high output (Benson et al. 2019; Lazear 2004). The Peter Principle is widely known among practitioners and has strong and consistent support from researchers. However, there has been no effort to extend this understanding to the broader area of Human Capital Strategy.

⁶"In no case is an animal activity to be interpreted in terms of higher psychological processes if it can be fairly

to either strategy or bias what can be better explained by measurement challenges. That is: the shape and tails of a worker's performance distribution are considerably more difficult, expensive, and time-consuming to measure than the mean. It follows that there is value in testing the role of worker volatility in a setting where performance measurement is unusually manageable.

2. Research Setting

Professional basketball has a long history as an empirical setting for questions about performance and randomness. Since Gilovich et al.'s (1985) foundational paper on shooting sequences, a large literature in psychology and sports analytics has used NBA data to study whether player output is more or less random than intuition suggests. Unlike that research, which uses the possession (really, the shot) as the unit of analysis, I examine performance at the level of a single game. One game gives a player ample opportunity to impact the outcome whereas a single possession can occur without the involvement of 9 out of 10 players in the game. Game performance has an advantage over season-level performance in that every game has a winner, a loser, and a common set of environmental and temporal conditions under which it is played. Still, this is a choice made (in part) for convenience. Volatility occurs from moment to moment (Dalal et al. 2020) and every period is the convolution of these micro-events. As our understanding of performance grows, it should naturally direct more attention to the scale at which workers are measured.

2.1. The Benefits of Professional Basketball as a Research Setting

Professional basketball offers three advantages for the study of worker performance and organizational outcomes. First, all organizations and workers are subject to a set of common rules and share (ostensibly) the same definition of success (Aversa et al. 2025). While there is certainly between-organization variation in resources, planning horizons, risk tolerance, and skill/sophistication, incentives broadly push all teams in the direction of winning games and championships within a reasonable time frame. Even cases of "tanking" or losing games in order to obtain roster-building

interpreted in terms of processes which stand lower in the scale of psychological evolution and development."

advantages in subsequent seasons are motivated by the eventual desire to win.

The second and third benefits derive from the fact that professional sports have rich and standardized performance data (Downs et al. 2025). Whereas many organizations are forced to hire, promote, or fire workers based on only a handful of observed data points, professional basketball teams can make decisions in an environment where every pass, shot, and step is logged. This greatly reduces the concern that decision-makers simply cannot see the distribution of a worker's performance. Were we to observe promotion decisions made in a less information-rich environment, we would first have to question whether or not the decision-maker was even aware of the full breadth of their employee's performance. And, as discussed, we would have every reason to think that they were not.

Finally, this wealth of data also allows researchers to partially observe the effort a player is exerting. In an oppositional game, more valuable actions are more costly. In the modern NBA, this means that the highest-value locations on the floor require the most energy to reach. A player seeking to shoot from 40 feet away can do so at any time (with the blessing of the opposing team). Higher-value shots (at the rim or from the free-throw line) require that the player be willing to work harder and, in all likelihood, absorb physical punishment in exchange for their shot attempt. Over the course of a season, a player's willingness to run themselves to exhaustion or be elbowed/shouldered somewhere soft will naturally ebb and flow, which is reflected in the type of shots they seek out.

2.2. Measuring Performance in Professional Basketball

Professional basketball teams play 82-game seasons. Each game lasts 48 minutes and the court is shared by five players on each team, giving 240 team-minutes per game. Teams take turns attempting to score baskets. An average game sees just over 100 possessions per team. On each possession, players can contribute to the success or failure of their team's attempt to score in multiple ways. Many of these are captured and tracked with basic counting statistics. In addition to points scored, players are measured on their shooting percentage (shots made divided by shots attempted), assists (passes which lead to teammate scores), blocks and steals (defensive actions

which prevent shooting opportunities for the other team), turnovers (possession lost without a shot attempt) and rebounds (ball recoveries after a missed shot).

As in many other jobs, basketball players have different strengths and inclinations, which frustrates attempts to collapse these statistics into a single number. Point guards, responsible for distributing the ball to others and chosen for their passing ability, will naturally get more assists than players who are on the court because they are tall enough to grab rebounds. The simplest approach, and the one I use in this paper, is a weighted sum of the player's activities called Game Score (Hollinger 2003). The idea behind Game Score is that each possession generates, on average, about one point. So an action which earns the team another possession is worth about a point while one that loses possession subtracts a point. These fractional "points" are then added to the player's point total, which gives their Game Score. The full formula is:

$$\begin{aligned}\text{Game Score} = & \text{Points Scored} + (0.4 \times \text{Field Goals Made}) - (0.7 \times \text{Field Goal Attempts}) \\ & - (0.4 \times (\text{Free Throw Attempts} - \text{Free Throws})) + (0.7 \times \text{Offensive Rebounds}) \\ & + (0.3 \times \text{Defensive Rebounds}) + \text{Steals} + (0.7 \times \text{Assists}) + (0.7 \times \text{Blocks}) \\ & - (0.4 \times \text{Personal Fouls}) - \text{Turnovers}.\end{aligned}$$

Game Score is not a perfect summary performance measure as it omits context (opponent quality, pace of play, location of game, quality of teammates, etc.). However, for the study of game-level performance variance it has several important features. Game Score is universally available, easily interpretable from game to game and has obvious face value. It penalizes bad things, rewards good things, and captures the many different ways to add value as a professional basketball player. A player whose Game Score falls from 30 to 5 in back-to-back games had one good game and one bad game, regardless of how they accumulated those statistics.

In order to ensure that the pieces of performance that are not measured in Game Score do not materially alter the below results, I also make use of one additional "advanced" statistic in the form of Box Plus Minus (BPM). This regression-based Game Score analog takes its name from

plus/minus, the team’s point differential while a player is on the floor, and estimates how much a player contributes to the team’s margin of victory, on average, across an entire season (Myers 2020). It includes much of the context that Game Score lacks and so is valuable for long-run and head-to-head comparisons. For individual games and the estimation of game-over-game performance volatility, BPM can be trickier to explain or interpret as it includes a more complex set of weights than those in Game Score. Where it is necessary to control for long-run player quality, I use BPM. Otherwise, I use BPM as a robustness check for Game Score based results.

The other problem with Game Score is that it is a counting statistic, so game-over-game variation is a function of both player characteristics and minutes played. This is akin to binomial noise. Consider, for example, that a single shot attempt has an average success rate of 46%. A player who takes 15 shots per game has a per-game sampling error of roughly 13 percentage points. At the league average of 9 shot attempts per game, this grows to roughly 16 percentage points (nearly three times the between-player standard deviation in shot percentage after correcting for this binomial noise). Prior work using NBA data (Awtrey et al. 2021; Barnes & Morgeson 2007) makes no correction or adjustment for this noise and so misspecifies the volatility of player performance.

Subsequent analyses use a minutes-adjusted Game Score standard deviation as an estimate of player-specific variance, net of the mechanical variance associated with limited playing time. For the median player-season (23 minutes per game), this correction removes roughly half of the observed variance. I first normalize Game Score per 36 minutes, which is conventional for NBA statistics and reflects the typical regular-season workload for a top player. This is necessary for comparison across players with different workloads, but it also magnifies the impact of variance. A particularly lucky 3 minute stint can look historically good (or bad) when multiplied by 12. I then decompose the variance of per-36 Game Score across games within a season into a player-specific intercept and accompanying noise:

$$\text{Var}_{\text{obs},i}(GS) = \underbrace{\sigma_{\text{true},i}^2}_{\text{player volatility}} + \underbrace{k \cdot \mathbb{E}[1/m_{i,g}]}_{\text{noise}}. \quad (1)$$

Here, $m_{i,g}$ is minutes played by player i in game g , and $k \cdot \mathbb{E}[1/m_{i,g}]$ is finite-exposure noise that shrinks as playing time grows. I estimate k by regressing observed variance on the player’s average inverse minutes for player-seasons in which a player plays more than 8 minutes in more than 20 games. The slope, $\hat{k}$, is approximately 650 (SE 21.7).⁷ Each player’s true variance is then his observed variance less this noise:

$$\hat{\sigma}_{\text{true},i}^2 = \max\{ \text{Var}_{\text{obs},i} - \hat{k} \mathbb{E}_i[1/m_{i,g}], 0 \}. \quad (2)$$

Hereafter, a player’s volatility refers to the square root of this adjusted variance.

Figure 1 demonstrates the importance of this correction. In the raw data, players who play less often look more volatile than those who play more minutes. If one had a prior belief that volatility was bad and steadier players should play more often, this would offer confirmation of that bias. After correcting for finite-sample noise, the relationship is roughly flat. If anything, the slight uptick in adjusted volatility suggests that the most important players are more volatile than those who are played sparingly.

Roughly 5% of player-seasons have observed variance below the estimated noise, given their minutes played. These players are so consistent that their game-over-game variation in performance is indistinguishable from the “bouncing ball” luck inherent to basketball. Figure 2 shows the spread of player-seasons from the sample period.

3. The Market Price of Reliability

3.1. Compensation and the Collective Bargaining Agreement

In the NBA, the rules governing on-court play and roster-building are negotiated between the league (representing the owners of the 30 franchise teams) and representatives for the players’

⁷The slope $\hat{k}$ is identified from cross-player variation and inherits any correlation between volatility and playing time. Within-player and composition-based estimates of the same slope are larger and so would remove some volatility which persists within player across seasons. The appendix shows results for alternate values of $\hat{k}$ and provides a split-sample estimate to gauge the magnitude of measurement error.

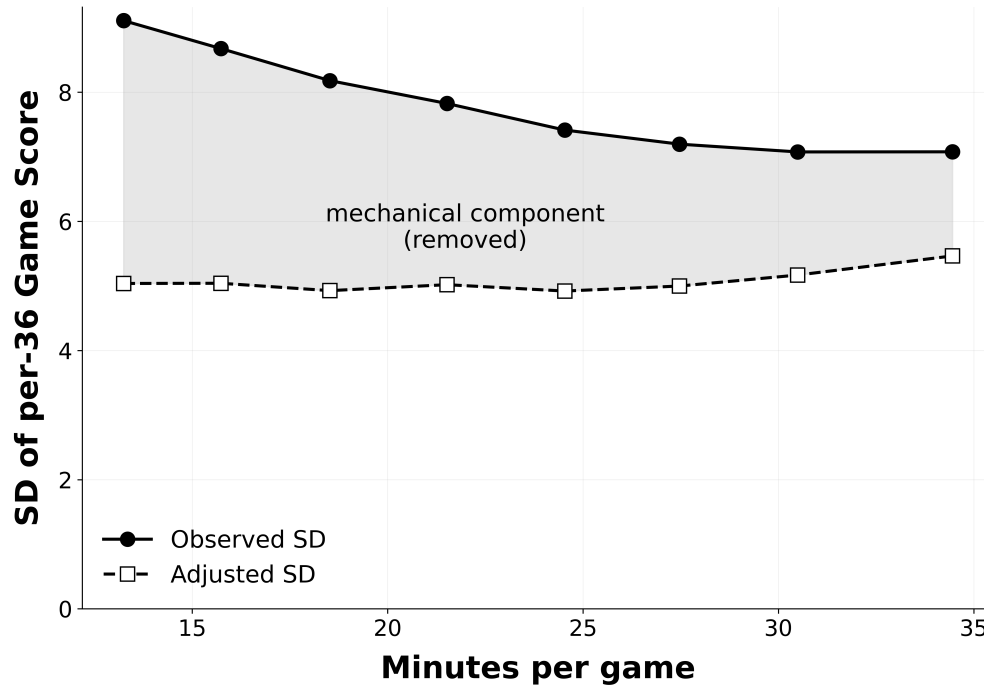


Figure 1: Observed and adjusted standard deviation of per-36 Game Score, by minutes per game. Points are bin means. The shaded region is the mechanical component removed by Equation (2). Chart shows player-seasons with at least twenty games of eight or more minutes, 2011–2023 ($n = 4,903$).

union (National Basketball Players Association or NBPA) as part of a collective bargaining agreement (CBA). Each CBA is an opportunity to substantially reshape the league landscape and the relationship between workers, fans, and teams. As such, the league’s history is typically broken up into eras defined by the prevailing CBA. This analysis is concerned with the “modern” NBA era, which began with the 2011 CBA and runs through the 2023–2024 season. The data set, drawn from Basketball Reference, includes 346,785 player-games.

The most important and contentious feature of a CBA negotiation is financial: the share of league revenue to be paid in salaries, the salary cap which all teams will adhere to, and all accompanying compensation rules. This includes the value, length, and payment terms of player contracts. There are three contract types of interest which are available to players in free-agency (able to negotiate compensation with multiple teams). First is the minimum contract, which ensures that no player will receive less than \$1.1 million (in 2023–2024) per year. Veteran players earn larger minimum salaries, subsidized by the league: a player with 10+ years in the NBA by 2024

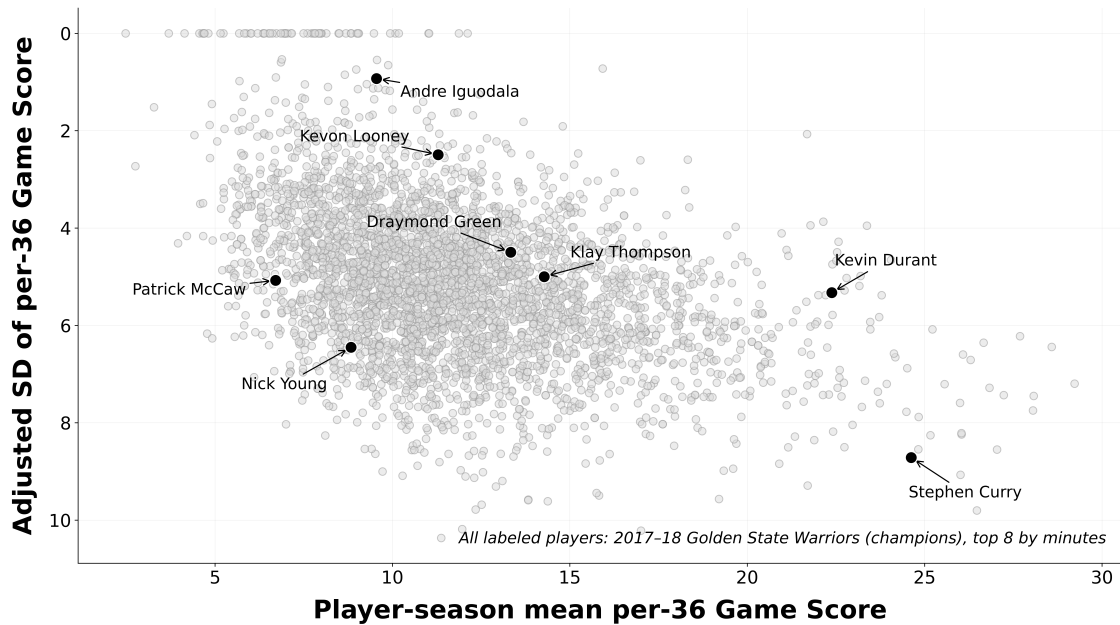


Figure 2: Scatter of player-seasons with 40+ games played from 2011-2023 ($N = 3,566$) by mean per-36 Game Score and adjusted volatility. Labeled points from the 2017-2018 Golden State Warriors.

could not be paid less than \$3.2 million per season. The minimum contract is also the only contract freely available⁸ to teams whose payroll goes over the salary cap (~\$136m in 2023-2024) and is generally reserved for players who are at or near the replacement level or quality floor of the league. While many minimum players provide good value to their teams, they are generally nearer to retirement or unemployment than to superstardom.

A second contract type is the Maximum contract, which pays a percentage of the salary cap. The specific percentage – 25, 30, or 35% (the “supermax”) – depends on the player’s years in the NBA and awards won in seasons prior to signing a new contract. The Maximum contract is almost exclusively reserved for the best players in the league (or players that teams project to become superstars), but it substantially underpays them. In 2013-2014, for example, Blake Griffin was paid just over \$16 million, but generated on-court performance that would have been valued, on an uncapped free-agent market, at around \$34 million.⁹

⁸There are various and limited exceptions to this rule, including the mid-level exception.

⁹Following FiveThirtyEight, I estimate a player’s open-market wage as the product of their wins (e.g., Win Shares) and the prevailing market price per win (approximately \$2.8 million in 2013–14).

Finally, players who are not offered a maximum contract but are too good to play on a veteran minimum negotiate wages in between the two extremes. The market for this “middle-tier” of free agents is, by no means unfettered or efficient. Teams are constrained from spending freely by the salary cap and their prior salary commitments. Rookie contracts are fixed by draft position and early-career players have their bargaining power reduced via “Restricted Free Agency” which gives their current team considerable leverage over the player and against other bidders. Players also value other compensation besides money. Some prefer to live in Los Angeles or to play with a superstar, beloved coach, close friend, or blood relative. Still, this is a substantial set of transactions in which the market values a player’s performance and the primary place in which a researcher can observe the wage impact of small changes in a player’s performance profile.

The NBA is often called a “superstar’s league” because team marketing, fan engagement, and competitive success tend to follow the best players. As such, teams are sophisticated in measuring and compensating top talent. A player whose expected performance justifies a maximum contract will likely have several suitors and the oddities of the CBA mean that the intensive margin of compensation with respect to performance is nonexistent. The same is true, albeit for opposite reasons, for players who are not good enough to earn above the minimum contract. If organizations value the distribution of a worker’s performance, it should show up in the middle tier of contracts alone.

3.2. Methods and Data

Across the data set, I observe 1,872 free-agent contract signing events and classify them according to the CBA’s wage scales.¹⁰ Of these, 119 are maximum contracts and 467 are minimum contracts (with both identified by the CBA’s published wage scales). The remaining 1,286 are middle tier. For clarity, these three tiers are subsequently subscripted as c_{cap} , c_{floor} and c_{mid} .

At each contract signing event, the primary dependent variable is the player’s log-transformed first-year salary on the newly signed contract. The independent variable is the player’s volatility

¹⁰Sample construction details in the appendix

(noise-adjusted Game Score SD) in a window of 1, 2, or 3 prior seasons (within-season variances pooled across seasons, weighted by games played). For each tier, I regress compensation on a measure of the player's prior-period performance volatility, controlling for prior-period mean performance and a battery of career-stage and role covariates (some shot profiles are inherently more volatile than others). The regression includes season fixed effects. Standard errors are clustered by player. Formally, for player i 's contract in time t :

$$y_{i,t} = \beta_{\sigma} Volatility_{i,t-1} + \beta_{\mu} Mean_{i,t-1} + Controls_{i,t} + \delta_t + \epsilon_{it} \quad (3)$$

Table 1: Variables and sample means by contract tier (signing events, 2011–2023)

Symbol	Definition	Mean by contract tier			
		Min	Mid	Max	All
y_{it}	First-year salary (\$M) of the newly signed contract; enters regressions in logs	1.43	6.67	25.46	6.56
$\text{Volatility}_{i,t-1}$	Noise-adjusted SD of per-36 Game Score over the evaluation window	4.77	5.10	5.72	5.10
$\text{Mean}_{i,t-1}$	Mean per-36 Game Score over the same window	9.81	11.48	16.93	11.68
age_{it}	Player age in signing year t	27.1	27.1	29.4	27.3
age_{it}^2	Squared age	—	—	—	—
$\text{tmin}_{i,t-1}$	Total minutes played, season $t - 1$; enters in logs	908	1,625	2,392	1,497
$\text{FGA}/\text{min}_{i,t-1}$	Field-goal attempts per minute, season $t - 1$	0.304	0.336	0.454	0.336
$\text{3PA}/\text{min}_{i,t-1}$	Three-point attempts per minute, season $t - 1$	0.104	0.102	0.115	0.104
$\text{FTA}/\text{min}_{i,t-1}$	Free-throw attempts per minute, season $t - 1$	0.074	0.087	0.148	0.088
δ_t	Signing-year fixed effects	—	—	—	—
N (signings)		467	1,286	119	1,872

The coefficients of interest are β_σ and β_μ . Both Table 2 and Table 3 report them in raw per-36 units: the change in log-wages associated with a one-point increase in per-36 mean Game Score or in the noise-adjusted per-36 standard deviation. Absent some bizarre circumstance, β_μ should always be positive, indicating that one additional point of Game Score is worth a higher salary. In raw units, the ratio $m_{mkt} = -\beta_\sigma/\beta_\mu$ represents the amount of mean Game Score a

team is implied to sacrifice, at a given wage point, in exchange for a one-point reduction in the noise-adjusted standard deviation of Game Score.

This analysis primarily examines the intensive margin, but volatility can also plausibly impact whether or not a player earns a contract of a given type or remains in the league at all. Using a multinomial logit model, I estimate the effect of increases to both mean performance and performance variance on the log-odds of a player's next free-agent contract falling into the middle or maximum tiers, rather than a minimum contract:

$$\ln \left[\frac{\Pr(\text{tier}_{i,t} = c)}{\Pr(\text{tier}_{i,t} = c_{\text{floor}})} \right] = \gamma_{\sigma,c} \text{Volatility}_{i,t-1} + \gamma_{\mu,c} \text{Mean}_{i,t-1} + \text{Controls}_{i,t} + \delta_{t,c}, \quad c \in \{c_{\text{mid}}, c_{\text{cap}}\} \quad (4)$$

I then use a simple linear probability model to test whether an at-risk player is re-employed. The sample keeps player-seasons of at least twenty games that are followed either by a new contract or by departure from the data set. Seasons whose continuation is contractually guaranteed are excluded, since those players mechanically cannot be forced to exit. This model uses the same set of controls and year fixed effects. Standard errors are clustered by player.

3.3. Results

For negotiated, middle-tier contracts, more volatile players earn less. Regression 1 evaluates log-wages on the prior season's game-over-game volatility without controlling for mean performance and recovers a positive coefficient, suggesting that more volatile players earn more than reliable players. As in other disciplines, the best NBA players are also players who play the most, do the most, and so have the largest spread on performance. However, the sign on β_{σ} reverses once the mean of the player is included in Regression 2. Over a single season, the penalty is statistically indistinguishable from zero (Regressions 2 and 3). It emerges, and grows monotonically, as the evaluation window is expanded to two and then three seasons (Regressions 4 and 5).¹¹

¹¹Sample sizes fall as the evaluation window widens because more prior-season data is required for the analysis.

A one-point increase in per-36 volatility (equivalent to +0.96 SD) measured over the prior three seasons is associated with a statistically significant 8.9% decrease in wages on the player's next free-agent contract ($p = .003$). When measured over two prior seasons, the penalty is 4.2% ($p = .07$). This trend reflects the difficulty of estimating a worker's variance compared to their mean (β_μ is precisely estimated in every evaluation window). As virtually all free-agent contracts are signed after 2+ (and usually 4+) years in the league, the coefficient in Regression 5 most accurately represents organizational attitudes towards worker volatility.

Volatility is persistent¹², rather than the artifact of a single season. When measured two seasons before the signing, or one season after it, a player's performance standard deviation predicts the wage penalty (Before: -0.042 , SE 0.012 ; and after: -0.038 , SE 0.014 , per point), and, obviously, post-signing volatility cannot have caused a contract that was already signed.

Table 2: Reliability premium on log(wage). Coefficients in raw per-36 Game-Score units; standard errors in parentheses, clustered by player. Columns (6) and (7) re-estimate the column (5) specification within the fixed tiers. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Negotiated Contracts					Fixed Contracts	
	(1)	(2)	(3)	(4)	(5)	(6) Min	(7) Max
β_σ (SD(GS))	+0.0345*** (0.0130)	-0.0059 (0.0135)	-0.0085 (0.0145)	-0.0432* (0.0235)	-0.0927*** (0.0313)	-0.0481* (0.0288)	-0.0146 (0.0180)
β_μ (Mean GS)	— —	+0.0696*** (0.0075)	+0.0707*** (0.0108)	+0.0807*** (0.0124)	+0.1005*** (0.0153)	+0.0193 (0.0149)	+0.0009 (0.0058)
age, age ²	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Log total minutes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Shot-attempt controls	No	No	Yes	Yes	Yes	Yes	Yes
Vol window (seasons)	1	1	1	2	3	3	3
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	970	970	970	785	581	95	75

¹²The appendix estimates roughly a 0.88 correlation in year-over-year trait persistence

The coefficients from Regression 5 are then used to generate estimates for the iso-value slope m_{mkt} , or the market’s tradeoff between mean performance and volatility. All three evaluation windows have a positive value, suggesting that wages trade off performance against consistency. However, the 95% confidence interval only excludes zero ([0.33, 1.52]) when teams have three seasons of prior performance to evaluate. Holding role, age, and shot profile constant, a value of 0.95 suggests that the market pays the same wage premium for a one-point increase in average performance as it does for a one-point reduction in volatility.

Table 3: Iso-value slope $m_{mkt} = -\beta_{\sigma}/\beta_{\mu}$, mid-tier signings. Raw Game-Score units; delta-method standard errors in parentheses; controls as in Table 2. * $p < 0.10$, *** $p < 0.01$.

Evaluation Window	β_{μ}	β_{σ}	m_{mkt}	95% CI	N
1 season	+0.0707*** (0.0108)	-0.0085 (0.0145)	+0.12 (0.20)	[-0.28, +0.52]	970
2 seasons	+0.0807*** (0.0124)	-0.0432* (0.0235)	+0.54 (0.29)	[-0.02, +1.10]	785
3 seasons	+0.1005*** (0.0153)	-0.0927*** (0.0313)	+0.92 (0.30)	[+0.33, +1.52]	581

Regressions 6 and 7 from Table 2 should return null results on the intensive margin. The minimum and maximum contracts offer pre-negotiated compensation, so a player who is already a max-worthy player does not earn more money by playing incrementally better nor is he punished for being more volatile. However, column (6) reflects partial-season minimum contracts,¹³ which suggests that volatile players, when signed, are likely to be signed later and paid only a pro-rated portion of their salary. This aligns with Table 4, which reports the results of the multinomial logit and linear probability models assessing the relationship between extensive margin and performance volatility. Column 1 reports the difference in odds of signing a mid-tier contract versus a minimum tier contract. Column 2 compares maximum and mid-tier contracts. Column 3 reports the likelihood that a free-agent is re-employed in the subsequent season.

¹³When we restrict the sample to full-season contracts only, the estimate is no longer significant.

Unsurprisingly, the results for mean performance are positive and significant. A one-point increase in mean per-36 Game Score raises the odds of signing a middle-tier rather than a minimum contract by 25% and the odds of a maximum rather than a middle-tier contract by 60% (both $p < .001$). However, the same increase in volatility has no significant effect.¹⁴

Volatility likewise does not push players out of the league. Among at-risk player-seasons, re-employment rises by 1.9 percentage points per point of mean performance (SE 0.6) and by 2.5 percentage points per point of volatility (SE 0.9), suggesting that teams may be more likely to give one last chance to a volatile talent who briefly shows some flash of high-level performance.

Table 4: Volatility and the extensive margin. Columns (1)–(2): odds ratios per one-point increase from Equation (4), p -values in parentheses (MLE standard errors; column (2) is the maximum-versus-middle contrast, delta method). Column (3) is the linear probability of re-employment for free-agents (player-clustered standard errors) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1) Mid vs. Min	(2) Max vs. Mid	(3) Re-employment
Mean (per-36 GS)	1.25*** ($< .001$)	1.60*** ($< .001$)	+0.019*** (0.006)
Volatility (SD per-36 GS)	1.05 (.511)	0.99 (.936)	+0.025*** (0.009)
Log total minutes	10.36*** ($< .001$)	9.89*** (.001)	+0.393*** (0.039)
Career stage (age, age ²)	Yes	Yes	Yes
Shot-attempt controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	1,244	1,244	1,870

This suggests that the results reported on the intensive margin are not the result of selection into the middle tier of contracts on the basis of a player’s performance spread. However, it is consistent with sequential performance evaluation. That is: teams initially assess how “good” a

¹⁴MLE standard errors may be biased when players sign multiple contracts within the sample period. However, a player-cluster bootstrap leaves the conclusions in the table unchanged.

player is based on their mean performance. For players who are not worth a maximum contract but desirable enough to justify a mid-tier contract, teams then price volatility in conjunction with mean performance.

It is important to note that the observed relationship is not causal. Thus far, I have made no effort to determine the compensation that a player would earn with their second-best offer or in some counterfactual in which they were more reliable but otherwise unchanged. Further, teams have more information about players than fans or researchers. Perhaps there are personality or environmental traits which make volatile players unattractive for reasons other than their on-court performance. Teams with detailed medical information and more insight into a player's precise tendencies may simply value dimensions of health, availability, or fit which are unavailable in the game/performance data.

While I remain unable to rule out the presence of privately known and unobserved (in this data set) heterogeneity, many of these confounders could plausibly bias the observed results towards zero. This first set of results suggests that teams will pay a premium for reliability. A trait that is widely known and disliked would already be priced into the player's wages. Teams with private information about some unattractive player characteristic will undervalue that player, relative to uninformed bidders, but they have no incentive to share that information with the 29 other franchises. And it only takes one stupid, desperate, or risk-loving team to create a market for the headcase that most other franchises would prefer not to employ.

4. The Organizational Value of Reliability

Though these results suggest that teams value reliability in the free agents they sign, they say nothing about the normative value of that reliability. Nor does the question "should teams pay for steadiness" have an obvious answer. On one hand, professional basketball teams are generally "smart" and so one might assume that the decision to pay for something is rational and well-informed. On the other hand, research on organizations is replete with examples of mispriced attributes, biased preferences, and an over-adherence to received wisdom. One need not look

further than professional sports, where baseball teams systematically overvalued batting average for generations (Hakes & Sauer 2006) and American football teams long resisted the statistically superior fourth down and two-point conversions (Hartley 2017; Romer 2006).

Even if we were convinced that streakiness is bad and consistency is good, it is possible that organizations have already evolved the tools and heuristics to manage it. In professional basketball, this might be the coach's scheme or playbook. In the broader set of organizations, this would be handled with routines and best practices. Of course, this may not even be necessary. After all, even compared to other professional sports, professional basketball allows for a large number of draws. A game has an average of 100 possessions (compared to 35-40 plate appearances per team in Major League Baseball) and a season has 82 games (compared to 17 in the National Football League and 38 in the English Premier League). And, unlike other sports, every player can contribute to every possession, so the law of large numbers has ample opportunity to insulate team outcomes from individual player volatility.

4.1. Methods and Data

Whether or not they are correct to price reliability, teams are undeniably strategic actors. They select their rosters and allocate playing time with clear objectives. In order to address the endogeneity of these decisions, I exploit the quasi-random timing of absences of key players. As a treatment, this has three useful features. First, while teams know that some players are more injury-prone than others, the precise timing of those injuries is difficult to predict. Further, teams use information about player injury risks strategically: fatigued players are given extra rest, prophylactic medical care, and tailored, state-of-the-art physical therapy, dietary advice, and equipment. Any injuries that result in spite of these measures are, I argue, an unpredictable residual. Finally, teams are heavily constrained in their mid-season roster choices, so absences almost always entail the reallocation of minutes to players who are already on the roster (and whose prior performance can be measured). In effect, each absence involves two pieces: the unexpected subtraction of a player and the promotion of their replacement. Both pieces are known and measured, so I compare the team's quality and

volatility before and after the absence against team outcomes. The estimand is the effect of the subtraction event as teams experience it, including whatever strategic response the team can muster. Results are substantially unchanged when directly controlling for the quality and volatility of the remaining roster.

I identify treatment events as player absences for which three conditions are met. First, the absent player must have averaged at least 15 minutes per game in the current season and played in each of his team's previous five games. This ensures that the absent players are active contributors (rather than experiments or fringe players) with sufficient performance history that I can examine pre-absence trends. Hereafter, I refer to these as "rotation players." Second, the player must play zero minutes for at least five games in a row.¹⁵ This is a long enough spell that the team must genuinely reorganize its rotation in response. Finally, I keep only the first qualifying spell of each team-season so that one absence event (and the corresponding roster/rotation changes) do not contaminate subsequent events. We might imagine that a team responds to an injury to its second-best player by asking the best player to do more. This added workload may cause a subsequent injury and make it impossible to disentangle the effects of the two absences from each other and from the impact of the shift in workload. By using only the first qualifying absence of the season, I minimize this attribution problem.

For each event, the focal player's mean and volatility are measured over the prior season, using noise-adjusted, mean-centered Game Score and BPM. Game Score remains the primary trait of interest, but BPM is useful as it is a more robust measure of player mean quality in the long run. Players without a prior season (rookies, players who missed an entire season due to injury, players who were under contract but not good enough to play) are excluded. Across the 2011-2023 seasons, there are 268 qualifying events. Absent players average 27 minutes per game before their absence with a prior-season per-36 mean Game Score of 9.0 and volatility of 5.1. The median spell

¹⁵I do not use formal injury designations for two reasons. First, the data available is incomplete, with some verified injuries being reported as "Did Not Play - Coach's Decision" and others not reported at all. Second, there is nothing unique about injury as a reason for a player's absence compared to paternity leave, bereavement, or suspension for cause. For my purposes, these all result in the unexpected subtraction of a player from the team's roster for an indeterminate amount of time.

lasts 11 games.

I estimate the treatment effect of losing a player using a two-stage, staggered treatment difference-in-differences estimator (Butts & Gardner 2022; Gardner 2022). A conventional two-way fixed-effect design may produce biased estimates when treatment is staggered because some already-treated observations are used as controls for later cohorts. This estimator avoids this by ensuring that treated games are excluded from the counterfactual. I use Gardner (2022) instead of Callaway and Sant’Anna (2021) because the parameters of interest are coefficients on continuous moderators of the treatment effect. In the first stage (identical to Borusyak et al. 2024), I estimate team-season and game number fixed effects by regressing the result of team j ’s game g as follows:

$$y_{j,g} = \text{TeamSeasonFE}_j + \text{GameNumberFE}_g + \epsilon_{j,g} \quad (5)$$

on $\{(j, g) : \text{UntreatedGame}_{j,g} = 1\}$

This first stage uses games before and after the treatment period. I drop the last game before the absence because injuries which occur mid-game produce distorted statistics. I then estimate the gap between predicted and actual team results for all games:

$$\hat{u}_{j,g} = y_{j,g} - \widehat{\text{TeamSeasonFE}}_j - \widehat{\text{GameNumberFE}}_g \quad \text{for all } (j, g). \quad (6)$$

In the second stage, I regress this gap against post-period indicators for player mean and standard deviation:

$$\hat{u}_{j,g} = \tau \text{Post}_{j,g} + \beta_\mu \text{Post}_{j,g} \text{Mean}_i + \beta_\sigma \text{Post}_{j,g} \text{SD}_i + \epsilon_{j,g}. \quad (7)$$

I use two outcome variables. One is a simple result indicator (1 if win, 0 if loss). A second is the point differential, where positive numbers indicate the margin of victory and negative numbers the margin of defeat.

Following Gardner (2022), which treats the two stages as a single estimation problem, I compute standard errors that add the uncertainty inherited from the first stage to the second-stage variance. I cluster at the team-season level because first-stage estimation error is shared by every

game within a team-season.

The coefficients of interest are τ , representing the average treatment effect on the treated (ATT), and β_μ and β_σ , which measure how the ATT varies with changes to the absent player's mean and volatility. Any claim to causality rests on a parallel-trends-style assumption. Here, that would mean that, absent the missing player, the treated team's results would have continued at the level implied by its own untreated games after adjusting for the usual rhythms of the NBA season. I discuss further below.

4.1.1. Treatment Timing

Figure 3 illustrates the construction of the pre-period, treatment period, and counterfactual samples. The first row illustrates a below-median length absence. The second shows a player whose absence is protracted. The game in which the player is injured or otherwise absented (region 2 in Figure 3) is not included in any samples because the player's statistics, in that game, are likely to be distorted by the mid-game injury.

I define the treatment-period as games 1-11 after the event or injury (region 3). This fixed window is designed to include the entire time period in which a team is adjusting to the absence of a key contributor. The median absence length is 11 games. I prefer a fixed post-treatment evaluation window length because a player's return from injury is partially a team choice. Some players are given long periods of time to recover to full health. Others are rushed back into action for a key game. Still others are traded and so never return to the team. Were the study to look at results only after the absent player returns, it would be capturing the effect of the team's timing decision.

The design estimates the team and time fixed effects from untreated periods of time (regions 1, 4, and 5), which is defined to include games before the player is injured/absent and after the treatment window. Though this approach eases concerns about the endogeneity of a player's return, it is biased in multiple ways. For absences that are shorter than the 11 game evaluation window, the "treatment" includes games where the absent player has already returned (region 3a in Figure 3). Games the player misses beyond the 11 game window (region 4) enter the counterfactual pool,

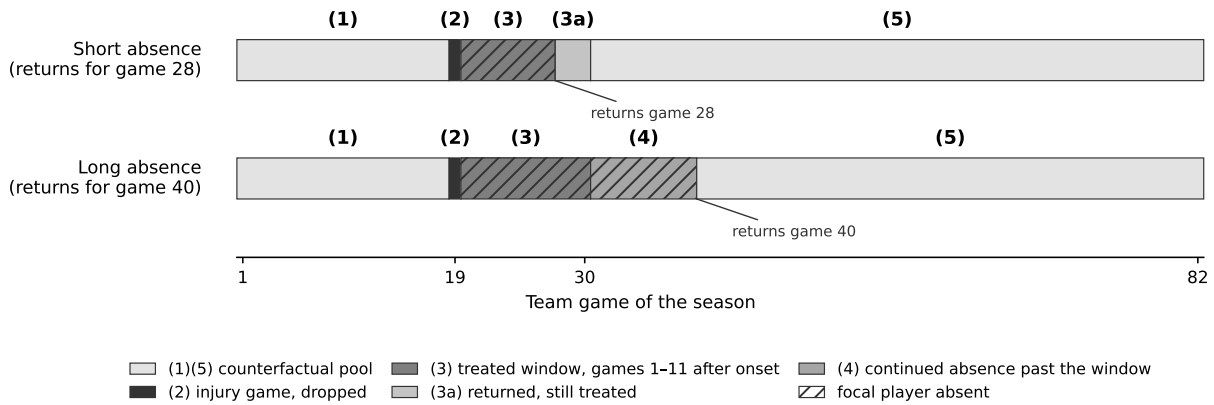


Figure 3: Illustration of treated and untreated periods for short and long injuries.

contaminating a baseline that is meant to represent the team with him available.

Because the treated event is the first qualifying absence of each team-season, no other qualifying absence is in progress when the window opens. However, when another rotation player misses games before the first returns (roughly half of the events overlap with another absence), I treat it as part of the effect being measured because subsequent injuries could plausibly have been caused by overexertion in response to the first absence. Fortunately, the biases introduced by both short and long absences work against the reported ATT.¹⁶ Games in Region 3a (player is available but assumed to be absent) reduce the observed impact of the absence. Games in Region 4 (player is absent but assumed to be available) will lower the baseline and so reduce the observed difference between counterfactual and treatment periods. The same logic also explains why I do not control for other absences to key rotation players outside the treatment window (though tests of the effect of non-focal player absences are given in the appendix).

4.2. Results

By using only absences to rotation players as the treatment, this study virtually guarantees that the average event hurts the team. A team plays its best players more than its worst players. When a player who receives significant playing time is absent, the team can only allocate those minutes to

¹⁶Appendix includes estimates from alternative specifications.

players that are worse (else they would have been playing in lieu of the lost player) or to players who are better but playing at capacity (else the team would already have allocated more minutes to them). Table 5 reports the expected negative and significant coefficient on the post-period indicator for both outcome variables. Any absence costs the team, on average, about a point per game and just over two percentage points of win probability.¹⁷

We would similarly expect a negative coefficient on β_μ . When a better player is absent, the team is hurt worse. Relative to the average absent rotation player, each additional point of prior-season Game Score costs the team about a sixth of a point of result margin and just over half of a percentage point of win probability. The positive sign on β_σ for all specifications suggests that an increase in the standard deviation of the absent player (holding mean performance constant) actually blunts the impact of their loss.

Table 5: Two-stage difference-in-differences estimates. Each row is a separate regression with different outcome variables. Two-step standard errors, clustered by team-season, in parentheses. $n \approx 21,500$ team-games; 268 treated cohorts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_μ	β_σ	τ	β_μ	β_σ
Point differential	-0.974*** (0.337)	-0.169* (0.090)	+0.612*** (0.189)	-0.974*** (0.334)	-0.319** (0.136)	+0.380** (0.154)
Win rate	-0.022* (0.012)	-0.0052* (0.0031)	+0.0222*** (0.0072)	-0.022* (0.012)	-0.0088* (0.0046)	+0.0124** (0.0055)

To illustrate the magnitude of these results, consider four hypothetical players (Table 6). Player A is the average absent rotation player in the sample, with a per-36 mean of 9 and volatility of 5. Because the moderators are mean-centered, the cost of losing him is about one point of margin per game, or a quarter of a win over the median eleven-game absence. Player B is a near-star whose absence clearly hurts the team. Player C is the steady performer whose absence is greatly missed.

¹⁷Because the moderators are mean-centered, τ is the effect for the average event and is identical by construction across Game Score and BPM.

Player D illustrates that sufficient volatility can completely offset the win-loss contributions of a league-average rotation player.

Table 6: Predicted cost of absence for four hypothetical players (Game Score coefficients, linear extrapolation).

	Mean (GS)	Volatility (GS SD)	Point differential per game	Losses (per-11 games)
A: average player	9	5	-1.0	+0.24
B: near-star	13	5	-1.7	+0.47
C: one point steadier	9	4	-1.6	+0.49
D: one point streakier	9	6	-0.4	0.00

4.2.1. Causality and Robustness

Figure 4 presents an event plot in which each bucket holds four games. In the four games before the absence and the first four afterwards, team point differential is statistically indistinguishable from zero (joint Wald $p = 0.49$ and $p = 0.16$ respectively). One possible explanation is that teams take immediate tactical and strategic steps to close the gap. However, the increased load on the remaining rotation (plus the opportunity for opposing teams to respond strategically) means that the relief is only temporary. In the subsequent 8 games, team performance falls significantly before rebounding in the final two buckets. The figure also splits the events into two cohorts above and below median volatility, suggesting that the ATT is largely driven by the absences of more reliable players.

The estimates in Table 5 support a causal story only if the games in the post-event evaluation window would otherwise have resembled the team’s season-long performance. More specifically, causality is threatened if there is something about the treatment window that impacts team performance, is correlated with absences, but does not result from player absences. This is different from other types of event studies which use the pre-treatment period to estimate the counterfactual and so must show that the post-treatment outcome was not already in progress when the event arrived.

I examine several plausible candidates. I first examine whether overexertion could be

responsible. If absences (particularly injuries) follow stretches of unsustainable, high-effort play, the subsequent results would be poor with or without absences. However, recent form has no consistent relationship with subsequent absences and the measured effects are unrelated to the team's pre-event performance (β_σ moves from 0.61 to 0.59 when an explicit pre-event-performance control is added). Likewise, there is no significant difference between treated and untreated games on location (home/away), opponent strength, or rest days between games. Finally, teams late in the season may change priorities (sometimes called tanking) from winning games to developing younger players or trading rotation players for other assets. The study design nearly rules this effect out by construction since only 5 of the 268 events begin at or after game 55.

Placebo tests using games before the event or after the absent player returns have negligible coefficients for margin of victory/defeat (pre-injury $\beta_\sigma = +0.08$, post-return $\beta_\sigma = +0.17$ for Game Score; +0.01 and +0.03 for BPM). Likewise, when the absent player's characteristics are replaced with those of a different, randomly selected player, estimates exceeding those in Table 5 are only reproduced in 20 out of 2,000 re-runs using BPM. For Game Score, it is 4 out of 2,000. Taken together, these robustness checks (reported in the appendix) suggest that whatever drives β_σ operates only while the player is not playing.

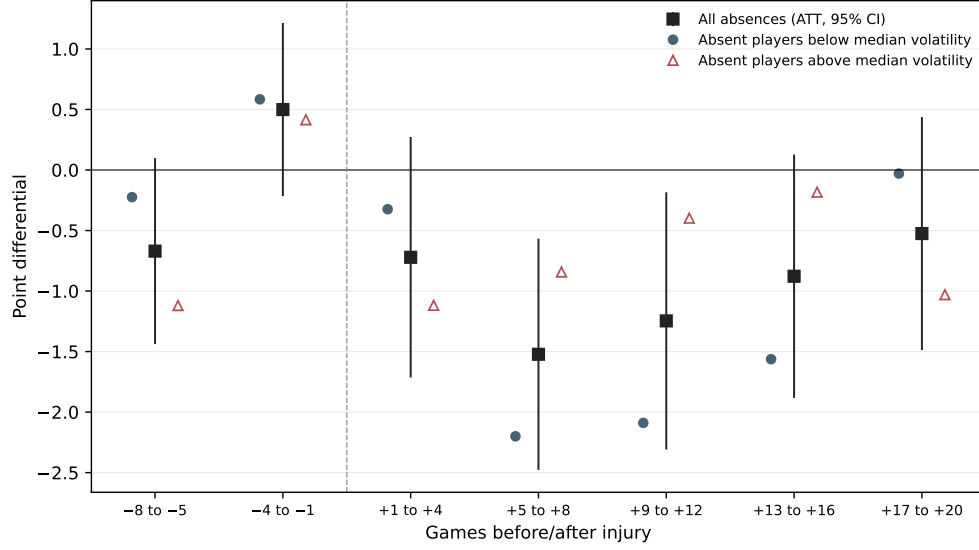


Figure 4: Team point differential around a rotation player's absence, in four-game buckets (the injury game, game 0, is omitted). 95% intervals from two-step standard errors, clustered by team-season.

Though I argue that the ATT is a causal estimate of the cost of losing a player, the moderators are not randomly assigned. β_σ and β_μ are descriptive statistics on the ATT, rather than causal statistics in their own right.

4.2.2. On Steadiness and Superstars

The results presented in Table 5 are largely driven by players who excel on one or more dimensions. Figure 5 reports the binned ATT by quintile of prior season mean or volatility. The bins are balanced on playing time. Where previous plots use Game Score, I now shift to using Box Plus Minus (BPM) because it is a more sophisticated measure when making player-to-player comparisons. BPM's game values are estimated at the season level and are not interpreted here. However, the game-to-game dispersion is at least as reliable a measure of spread as Game Score's. Once more, negative values indicate the estimated harmful impact to team performance from losing a player with mean or standard deviation in a given quintile. The results for mean are intuitive. Losing below average players does not hurt the team nearly as much as losing a very good player (BPM above 4.0 is considered "All-Star Quality" per Myers (2020)).

Likewise, the loss of one of the most reliable players hurts team point differential. What is somewhat surprising is the fact that the magnitudes of the two effects are somewhat similar. Losing a star player costs the team 2.64 points per game. Losing a very reliable player costs 2.14 points.

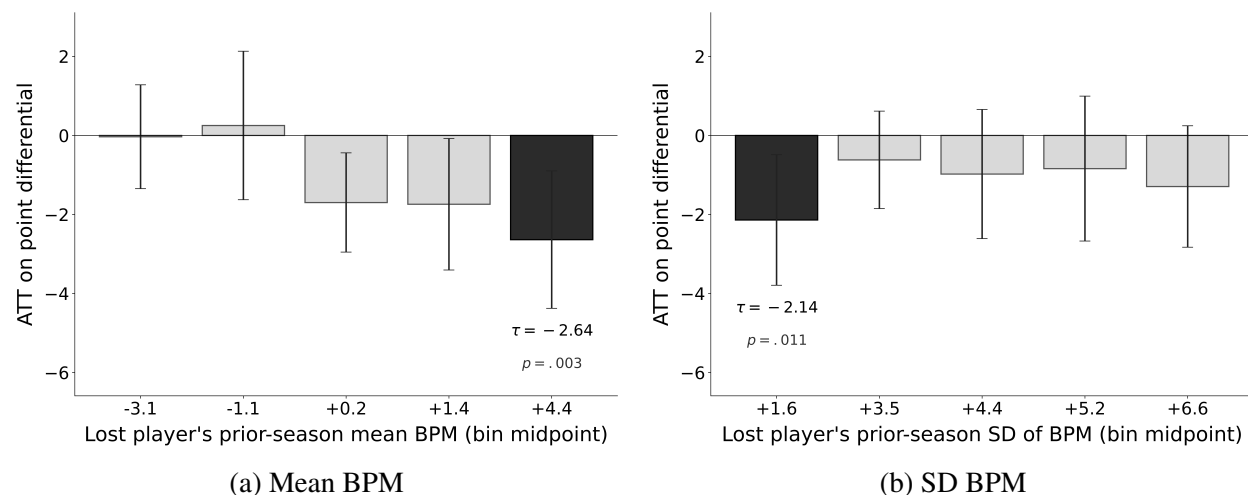


Figure 5: Post-injury ATT on point differential by minutes-balanced quintile of the absent player's mean or SD of prior-season BPM. Error bars are $\pm 1.96 \cdot$ bootstrap SE.

Interestingly, these appear to be almost completely different populations of players. Figure 6 plots the observed player absences and highlights the players in each of the top quintiles from Figure 5. The overlap is tiny. There are only four players (out of 268) with observed absences who are both all-star quality and highly reliable. Of course, this is not to say that players cannot be both. Rather, the rare player who is a highly reliable superstar is also unlikely to get injured early in the season. Still, evidence suggests that Reliable (triangles) and High-Average (squares) are two different populations with roughly similar marginal replacement costs. Crucially, the former delivers 80% of the impact while being paid, on average, 40% as much as the latter group (\$18.7M per year versus \$7.5M).

5. Discussion

The wage study, which shows that organizations do place a premium on reliable workers, has been done before with conflicting results. Barnes and Morgeson (2007) suggest that there is a negative

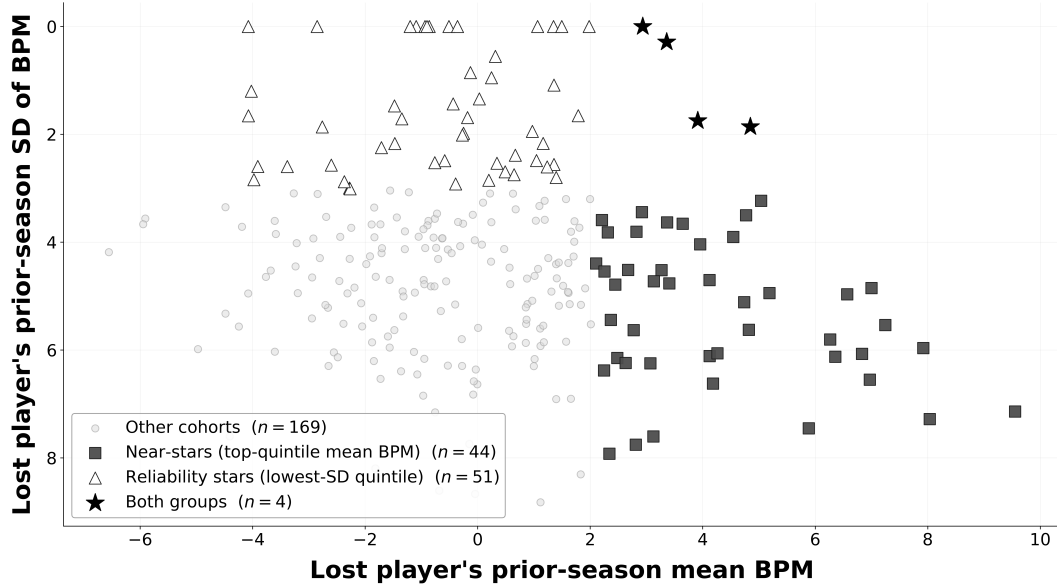


Figure 6: Absent or injured players by prior-season mean BPM and prior-season SD of BPM. Triangles represent players in the top quintile of Reliability. Squares are the top quintile of mean performance. Stars are in both categories.

association between performance variability and wages. Awtrey et al. (2021) find that organizations do not pay an appropriate premium for reliability. Neither analysis separates persistent, player-specific volatility from variation generated by playing time nor allows for the compensation nuances in the NBA CBA. After adjusting for these measurement and institutional constraints, I show a positive, significant relationship between a player's game-over-game reliability and their negotiated compensation. The association also emerges more strongly as teams accumulate multiple seasons of performance information, consistent with the greater difficulty of estimating volatility than mean performance.

The second analysis suggests that volatility is genuinely detrimental to organizational performance and outcomes. Even after the best efforts of coaches, managers, and teammates, who have more information about players' roles and tendencies and the ability to adjust schemes and rotations strategically, there is still a persistent and negative effect when a reliable cornerstone of the team is lost to injury. This validates the finding that teams price reliability as a distinct player performance characteristic. The coefficients in Table 3 are consistent with those in Table 5. Organizations trade roughly one point of mean performance for a one-point reduction in Game

Score standard deviation. They could reasonably value that same reduction at 3.6 points of mean production, though the ratio is imprecise (95% CI -0.5 to $+7.8$).

In the real world, superstars are rare. Not every company can have one, let alone an army of 10x engineers or A+ sales reps. And for companies with limited resources, top-tier human capital is often out of reach on the labor market. These results suggest that those organizations may find that it is better to build a steady and consistent, if unspectacular, workforce, rather than compete with the deep pockets of the world's largest and most prestigious firms. This is not to say that superstars are not important (most of the aforementioned highly reliable players share a roster with one or more max-contract superstars), but for companies who are constrained relative to their competitors in the market for talent, reliability has a quality all its own.

Missing, from this analysis, is more detail on the exact mechanism that drives the results. After all, I have shown that players who consistently score fewer points (or do fewer things) can be, on average, better for the team than those that score in bunches with fallow periods in between, which is, perhaps, counterintuitive. As this is a paper about organizations, performance, and strategy and not a paper about professional basketball, the specific basketball mechanisms are largely outside the scope of the project. However, I can note three mechanical features that generalize beyond professional basketball.

First, success and failure are binary and separated by a knife's edge. The first point that the opponent cannot match is extremely valuable (the team wins by 1) but the second point of win-margin is worthless (a win by two points is the same as a win by one point).¹⁸ In effect, the team is forbidden from keeping any excess performance past some threshold. This is a common feature of organizational payoff functions and suggests areas where reliability or safety should be valued: Political parties need only win by one vote. Entrants need only move fast enough to erect entry barriers behind them. And startups need only enough revenue to pay the bills in order to live to fight another day. In environments with the opposite incentives (limited or fixed downside, lots of upside), we would expect to see a premium on volatility (venture capital investing,

¹⁸Though professional basketball has flirted with rule changes which would modify this, including the Elam ending's target-score finish.

computational/algorithm races).

Second is the shape of worker performance itself. Game-level output is right-skewed for roughly 90% of the rotation players in professional basketball. A player's best decile of games averages 1.9 standard deviations above his own mean, while his worst decile averages only 1.5 below.¹⁹ Volatile players, then, do not trade good games for bad games, one for one. They are trading many games below average for a handful that are well above average, with team payoffs again, limited by the nature of binary win-loss outcome.

Finally, as is true in many organizations, basketball is a team sport in which a player's contributions affect both the scoreboard and the contributions of their peers. A reliable teammate, in basketball, helps the offensive and defensive schemes function even when other players are out of position. They provide insurance that covers for mistakes or allows others to take risks without fear of outsized negative consequences. In settings where performance is entirely independent, we should see a weaker preference for reliability (as in professional baseball, which values power hitters highly, even if they strike out often). And where teamwork matters, it is worth paying a premium for consistency.

6. Conclusion

This paper suggests that organizations do price worker volatility consistent with its impact on organizational outcomes. Where the CBA allows prices to move, professional basketball teams pay a premium for player consistency. As evidence of a player's volatility accumulates, teams are more likely to reward consistent players and pay less for the same mean performance from volatile players. When an event removes a worker, the cost of his absence rises with his steadiness (holding average performance fixed) and is not inconsistent with the wage the market pays.

In strategic factor markets, firms can capture value when they have more accurate expectations than competitors about the future value of a resource or the evolution of the world state (Barney 1986), which makes the generation, gathering, and interpretation of information into im-

¹⁹Player-seasons with at least forty appearances and fifteen minutes per game, 2011–2023.

portant capabilities for sustained competitive success (Makadok & Barney 2001; Ryall & Sorenson 2025). Though people are the eponymous human resource, strategic human capital would benefit from developing on these well-understood ideas. After all, worker reliability is important to organizational success, difficult to know ex-ante, costly to estimate, and slow to be reflected in labor-market prices. Organizations that measure it more accurately may have a substantial and sustained advantage over their competition.

Obviously, organizations that excel at identifying or manufacturing reliability can sustain this advantage over an extended period of time. However, there is also evidence that workers' reliability is not fixed. When an individual's penchant for good or bad days evolves, it lengthens the window of time in which their productivity will be mispriced. In effect, the labor market can be precise about a worker's volatility or current, but not both. An organization that observes its workers daily and so can more quickly and accurately gauge the second moment of the performance distribution has no such tradeoff. As the worker evolves, the organization's advantage is renewed.

Finally, I return to the question of why volatility is so conspicuously absent from the literature on human capital strategy and organizational design. After all, practitioners and researchers are aware of the Peter Principle. Why hasn't research extended the concept further?

This paper shows that at least one type of organization does price worker volatility where it is observed, so it's difficult to argue that decision-makers and researchers have a universal blind spot or bias. And even in a setting with ample strategic behavior and substantial scale (100 possessions per game, 82 games per season, each with 10 workers participating), the normative effect remains. Losing someone consistent produces an outsized and negative effect on team performance. This is not to say that buffering or formal routines and structures are irrelevant, but it does suggest that they are incomplete and costly enough to present organizations with a dilemma. Likewise, the law of large numbers cannot fully average away outlier performances in a competitive setting with fine margins.

What remains is measurement. While there is substantial evidence that volatility is a semi-stable worker characteristic, there is widespread disagreement about the shape of the distribution

from which performance is drawn (Beck et al. 2014; O’Boyle Jr. & Aguinis 2012). Without knowing whether a worker’s performance is normal, log-normal, or heavy-tailed, how much data must an employer collect before confidently calling that worker reliable or volatile? And for researchers, attempting to simultaneously estimate both moments of each worker’s performance distribution is likely to strain most field and lab studies. Even in this setting, more than half of observed variance is mechanical noise and the wage penalty is only visible in the data over multi-season evaluation windows. But, where measurement is available, behavior consistent with rational and strategic pricing follows.

Research on human resources should, therefore, either have a credible reason for reducing workers to the mean or take steps to ensure that interventions which raise the average do not materially and adversely impact the distribution of worker performance. Measurement, in turn, gives way to management, which is important because workers do not produce averages. They deliver noisy, messy sequences of good and bad days, and the organization gets to experience them all.

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Appendix to *Playing the Spread: Worker Reliability and Organizational Performance*

APPENDIX A: LLM/AI USAGE

I used LLM tools in three ways for this project. First, I managed my code through a Claude Code interface using mostly Opus (with some work done by other models as tokens/availability permitted). I also used Claude to directly generate Figure 3 (all other charts and figures are Python code outputs). Finally, I used Claude Fable, GPT 5.6 Sol, and Refine.ink to proofread the manuscript and appendix. All writing is my own. So, too, is the theory and the research design.

APPENDIX B: SAMPLE CONSTRUCTION

Middle-tier free-agency signing events are identified by the percentage change in annual value. Contractual raises cannot exceed 8%, so any larger increases must be achieved in free agency. I also include salary decreases of the same magnitude. This approach excludes players who re-signed at essentially unchanged salaries. The transaction logs that I have access to are substantially incomplete, so the use of salary rules provides a larger set of events. However, the logs do allow me to identify an additional set of 186 free-agent signing events at essentially unchanged wages, which I also include in the sample.

It is possible that there are other events of this type which are omitted from my sample. Table B.1 suggests, however, that the inclusion of these additional 186 free-agent signings increases both the magnitude and significance of the coefficients.

Table B.1: Estimates with and without same-wage re-signing events. Coefficients on volatility in raw per-36 Game Score units. Standard errors in parentheses, clustered by player. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

	1 season	2 seasons	3 seasons
(1) Main specification	-0.0085	-0.0432*	-0.0927***
	(0.0145)	(0.0235)	(0.0313)
Signings	970	785	581
(2) Excluding same-wage re-signings	-0.0098	-0.0339	-0.0773**
	(0.0155)	(0.0241)	(0.0309)
Signings	886	716	520

APPENDIX C: ESTIMATING PLAYER VOLATILITY AS A CHARACTERISTIC

A player's observable game-over-game volatility is closely related to the amount that they play. Players who see the court less in each game have a higher game-over-game variance, so an accurate estimate of player volatility depends on being able to separate noise associated with short (or long) playing stints from persistent, player-specific volatility. The parameter k determines how much to weight playing time in the variance decomposition. In the main analysis, I estimate $k = 650$ from 4,903 player-seasons in which a player plays 20 or more games with 8 or more minutes. When applied to all player-seasons (5,431) of at least 10 games and 8 minutes played, this removes a median 53% share of observed variance and suggests that roughly 5% of player-seasons are so consistent in their performance that we cannot separate individual volatility from chance. The median player-season standard deviation of per-36 Game Score falls from 7.58 to 5.17.

Table C.1 shows a series of additional values for k and their impact on the sample statistics. Larger values of k result when estimating the impact of differences in playing time using only within-player minutes variation. Subsequent rows further attempt to remove sources of noise for a given player by controlling for the number and type of shots he takes (the largest source of noise in the Game Score statistic). The last row abandons regression entirely and instead

estimates k_i for each individual player from his own mix of shots (twos, threes, free-throws) and same-season shooting performance.

Table C.1: Alternate noise slopes k and their consequences for the volatility measure.

k	Estimated from	Share of variance removed	Players with zero volatility	Median Adjusted SD	Corr(SD_t, SD_{t+1})
0	no correction (raw SD)	0%	0.0%	7.58	0.35
650	cross-player regression (used in the paper)	53%	5.1%	5.17	0.33
999	within-player minutes variation	81%	28.3%	3.30	0.39
1,051	usage-adjusted, at mean shot volume	85%	32.7%	2.92	0.39
1,066	shot-mix composition, median player	86%	34.0%	2.81	0.39
k_i	shot-mix composition, own slope per player	86%	26.2%	2.79	0.05

Table C.2 shows how variation in k impacts the main results shown in Table 2. That is: middle-tier free-agent salaries with three years of prior playing history. The coefficient β_σ shrinks, as expected, when more variation in variance is removed from the sample. However, it remains significant for all except the player-specific $k = k_i$.

Table C.2: Three-season wage coefficients for different values of k . $N = 581$. Standard Errors clustered by player.

k	β_μ	(SE)	β_σ	(SE)
0	+0.0971***	(0.0146)	-0.1575***	(0.0368)
650	+0.1005***	(0.0153)	-0.0927***	(0.0313)
999	+0.0986***	(0.0155)	-0.0488**	(0.0229)
1,051	+0.0972***	(0.0155)	-0.0397*	(0.0224)
1,066	+0.0969***	(0.0155)	-0.0377*	(0.0223)
player-specific	+0.0901***	(0.0147)	-0.0108	(0.0236)

As an additional check to ensure that the adjusted standard deviation reflects a worker-specific feature, I split each player-season into two halves (odd and even games) and compute the adjusted volatility separately for each half. Being drawn from a single player, the standard deviation in each half-season is a noisy estimate of the same characteristic. I then use one half as an instrument for the other. The wage regression itself is unchanged. Because the two halves share no games, this specification removes independent, game-over-game luck. Table C.3 compares the coefficients found in Table 2 (the OLS column) to the corresponding IV estimates.

On a single year of prior performance, the instrumented effect is positive and insignificant ($F = 15$). With two seasons of data, the magnitude of the IV estimate is comparable to the OLS estimate, but not statistically significant. However, the three-season evaluation window is a strong instrument ($F = 39$) and shows a larger (and still significant) effect compared to the OLS estimates. One odd feature of this approach is that teams observe (and choose wages) based on the full sample of games. So if they are sophisticated in attempting to parse out a worker's volatility, the IV coefficients are more representative than the OLS coefficients. If, instead, they are mostly reacting to observable behavior with a basic correction for minutes played, the true value is likely closer to the OLS estimates.

Table C.3: Split-sample instrumental-variable estimates of the wage response to volatility, on the estimation sample of Table 2. Coefficients are per SD point of volatility.

Window	OLS	(SE)	IV	(SE)	F	Split-half r	N
1 season	-0.0085	(0.0145)	+0.0712	(0.0852)	15	0.35	970
2 seasons	-0.0432*	(0.0235)	-0.0521	(0.0828)	30	0.44	785
3 seasons	-0.0927***	(0.0313)	-0.1574**	(0.0712)	39	0.51	581

Table C.4 tests the sensitivity of the results in Table 5 to changes in k . As before, the coefficient on β_σ decreases as more variance is pruned away but the estimates remain significant and material. Note that because the moderator's scale shrinks as k grows, the second pair of columns restates the coefficient per one standard deviation. With this normalization, the coefficients are largely invariant to the choice of k .

Table C.4: Estimates from the difference-in-differences design using injuries as treatment and volatility as a moderator (β_σ) for each alternate noise slope k . Moderators are measured in Game Score and the outcome variable is point differential. $N = 268$.

k	β_σ (per point)	(SE)	β_σ (per SD)	(SE)
0	+0.6705**	(0.2689)	+0.739**	(0.296)
650	+0.6122***	(0.1888)	+0.949***	(0.293)
999	+0.4570***	(0.1502)	+0.960***	(0.315)
1,051	+0.4606***	(0.1522)	+0.983***	(0.325)
1,066	+0.4627***	(0.1520)	+0.990***	(0.325)
player-specific	+0.3427**	(0.1649)	+0.576**	(0.277)

The choice of k is, ultimately, a judgment call. I have elected to estimate k from all players because it is intended to remove only the variation associated with minutes played. The objective is to estimate the persistent, individual *characteristic* and I believe it is safer (and more informative) to avoid controlling too much for observable differences in players. After all, those differences are what the market prices and the organization derives value from.

C.1 Volatility as a Stable Trait

How stable is volatility over time? As shown in Table C.5, the season-over-season correlation in player-specific volatility is 0.33. Rows (2) and (3) then look at the between-year correlation for gaps of two and three years. Row (4) repeats the split-sample (odd and even numbered games) and provides the within-season correlation.

First, note that volatility is substantially less “stable” than a worker’s mean performance, which is to be expected given the relative difficulty in measuring the former compared to the latter. Both traits drift over time. This is consistent with the literature on learning curves, which suggests that workers grow more skilled and more reliable as they gain experience.

Table C.5: Within-career drift of the player’s mean and volatility. Cells are correlations across the measure-construction frame of Table C.1 (5,431 player-seasons, 2011–2023, per-36 Game Score).

	Volatility	Mean performance
(1) Correlation year-over-year	0.33	0.80
(2) Correlation, two years apart	0.29	0.74
(3) Correlation, three years apart	0.27	0.68
(4) Correlation, odd and even games	0.26	0.78

Dividing the two-year correlation by the year-over-year correlation (in an effort to separate sampling noise from persistent trait) suggests that the year-over-year persistence in the player’s individual volatility is as high as 0.88.

APPENDIX D: SELECTION AND SURVIVAL

More performance history is useful in estimating a player’s personal volatility. However, many players play fewer than four seasons (focal season plus three prior seasons as used in Table 2) and are subsequently not offered a contract. This is partially addressed by the analysis of the extensive margin, but as a robustness check, it is useful to ensure that the significant coefficient on β_σ with three prior years of data is not driven by selection.

I re-estimate the wage regressions using only the 581 free-agent signings for which there is a full three years of prior performance data in the sample. As in the manuscript’s main specification, a single season of performance is insufficient. As more data becomes available, the coefficient becomes larger and more significant, suggesting that the manuscript’s results are not an artifact of changes to the sample population.

Table D.1: The reliability premium at one, two, and three seasons of prior performance. All-signings cells reproduce Table 2, columns (3)–(5), whose samples shrink as the window lengthens. Common-sample cells hold the sample fixed at the 581 signings with three seasons of prior performance. Coefficients on volatility in raw per-36 Game Score units; standard errors in parentheses, clustered by player. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	1 season		2 seasons		3 seasons	
	All	Common	All	Common	All	Common
Prior performance						
β_σ (SD(GS))	−0.0085 (0.0145)	−0.0109 (0.0189)	−0.0432* (0.0235)	−0.0669** (0.0278)	−0.0927*** (0.0313)	−0.0927*** (0.0313)
Signings	970	581	785	581	581	581

APPENDIX E: ROBUSTNESS OF THE UNTREATED GAME DEFINITION

The manuscript uses a fixed, 11-game window post-absence as the treatment period in order to minimize the impact of strategic roster management. Teams have considerable discretion in when they bring players back from injury. An organization which desperately needs to win games may return a player to the lineup as soon as possible. One without that imperative may wait until they are fully healthy. However, the use of a fixed window allows untreated games to intrude into the treated window and treated games to fall outside the window. I have argued that both are likely to bias results to zero. Table E.1 replicates the results shown in Table 5 using a variety of different definitions of “untreated” games. Specification (1) is the standard, 11-game treatment window in which all games before the absence and starting 12 games afterwards are used along with never-treated games from other teams to construct the baseline. Specification (2) removes all

consecutive games in which the injured player is absent and (3) further removes all games in which any important player is in the midst of a 5+ game absence. Finally, Specification (4) estimates the untreated level only from pre-treatment games and never-treated team-seasons.

As the criteria for untreated games become more restrictive, the coefficient for the ATT (that is, the average cost of losing a player) grows substantially. Though more restrictive definitions lower statistical power (Specification 4's untreated pool contains less than one third of the games used for Specification 1), the coefficient on volatility remains positive. β_{σ} for Game Score only becomes statistically insignificant in the win rate panel when using the most restrictive Specification 4. The Box Plus Minus volatility moderator, which is generally the less significant coefficient, loses significance for specifications (3) and (4), but remains positive. The consistency of these signs suggests that the results reported in the manuscript are not particularly sensitive to the games which are used to construct the untreated baseline.

Table E.1: Alternate definitions of untreated games. The four specifications use 18,608, 16,015, 8,774, and 5,458 games, respectively. Two-step standard errors, clustered by team-season, in parentheses; 268 events in every row. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_{μ}	β_{σ}	τ	β_{μ}	β_{σ}
<i>Point differential</i>						
(1) All games outside the treatment window (main text)	-0.974*** (0.337)	-0.169* (0.090)	+0.612*** (0.189)	-0.974*** (0.334)	-0.319** (0.136)	+0.380** (0.154)
(2) Also excluding the rest of the focal player's absence	-1.015*** (0.350)	-0.224*** (0.082)	+0.577*** (0.191)	-1.015*** (0.346)	-0.429*** (0.119)	+0.315** (0.157)
(3) Also excluding all qualifying absences	-1.416*** (0.392)	-0.290*** (0.092)	+0.633*** (0.216)	-1.416*** (0.390)	-0.483*** (0.137)	+0.299* (0.174)
(4) Pre-absence and never-treated games only	-1.446*** (0.498)	-0.178* (0.099)	+0.402* (0.243)	-1.446*** (0.495)	-0.374*** (0.145)	+0.221 (0.190)
<i>Win rate</i>						
(1) All games outside the treatment window (main text)	-0.022* (0.012)	-0.0052* (0.0031)	+0.0222*** (0.0072)	-0.022* (0.012)	-0.0088* (0.0046)	+0.0124** (0.0055)
(2) Also excluding the rest of the focal player's absence	-0.024* (0.012)	-0.0064** (0.0030)	+0.0222*** (0.0074)	-0.024** (0.012)	-0.0113*** (0.0044)	+0.0104* (0.0057)
(3) Also excluding all qualifying absences	-0.037*** (0.014)	-0.0075** (0.0034)	+0.0170** (0.0084)	-0.037*** (0.014)	-0.0130*** (0.0050)	+0.0063 (0.0063)
(4) Pre-absence and never-treated games only	-0.038** (0.017)	-0.0050 (0.0036)	+0.0123 (0.0091)	-0.038** (0.017)	-0.0116** (0.0053)	+0.0062 (0.0068)

APPENDIX F: ABSENCE TIMING

F.1 Strength of Schedule and Tanking

In the absence-based event study, the game-of-season and team-season fixed effects capture much of the temporal variation in the sample. However, it is possible that absences correspond to particularly challenging games. Because all wins count the same, a team might choose to rest players who are carrying nagging injuries (or are simply fatigued) during a stretch of difficult games. Though this is unlikely to impact the reported results (5+ games would be an unusually long absence for load management), I additionally test whether absences (really, the 11-game windows following an absence) correspond to more challenging stretches of the schedule.

I estimate a team's likelihood of winning each game before the season begins using both teams' prior-season points for and against and an indicator for game location (home or away). I then combine this estimated win probability with the game's actual result to estimate each team's win rate in excess of expectation. This excess win rate is then used as the dependent variable in the same regressions shown in Table 5. As an alternate test, I re-estimate the main results without the five events that occurred late in the season, which are the most likely candidates for tanking (losing games in order to gain benefits in subsequent seasons).

The results in Table F.1 closely match those for Win Rate in the manuscript results table. Additionally, there is no material relationship between schedule difficulty and absence timing. Treated games are, on average, 0.1 percentage points more winnable than untreated games (SE = 0.3 percentage points), which is negligible.

Table F.1: The Table 5 specification re-estimated with win rate in excess of expectation as the outcome variable (first row) and dropping late-season events (second row). Two-step standard errors, clustered by team-season, in parentheses. 268 events in the first row and 263 in the second. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_μ	β_σ	τ	β_μ	β_σ
Win rate in excess of expectation	-0.023** (0.012)	-0.0053* (0.0030)	+0.0208*** (0.0070)	-0.023** (0.012)	-0.0093** (0.0045)	+0.0123** (0.0053)
Win rate, excluding events at game 55 or later	-0.022* (0.012)	-0.0044 (0.0031)	+0.0224*** (0.0072)	-0.022* (0.012)	-0.0079* (0.0047)	+0.0130** (0.0055)

F.2 Reversion to the mean

I further argue that absences are quasi-random. Even knowing that a player is injury prone (or likely to be absent for other reasons, including paternity leave or suspension), teams cannot predict the exact point in time at which they are likely to become injured in advance. However, injuries, which represent the majority of absences, can be caused by overexertion. Overexertion can also result in unsustainable positive results before the injury. So what if the observed treatment results are, in fact, regression to the mean, helped along by a key absence? Alternatively, we could also imagine that injuries happen when teams are already trending downwards, either because a player is carrying some small injury which inhibits their performance or due to other, non-injury related factors.

Table F.2 shows the results of a linear probability model in which the team's form (average point differential) in the prior 3, 5, and 10 games predicts an absence event. Positive values mean that prior over-performance predicts an absence and are consistent with the idea that overexertion is partially to blame for the reported results.

The analysis is inconclusive. The prior three games have a positive but insignificant coefficient. When the window expands to five games, the coefficient becomes significant but

remains small: a five-game period of over-performance increases the likelihood of absence by roughly 0.11 percentage points per added point of form. However, when the window is further expanded to 10 games, the sign flips and is, again, insignificant. It is tenuous to suggest that a positive performance driven by effort alone has a delayed effect on the player's injury likelihood, but we could contrive a story around elevated effort with diminishing returns over time.

Table F.2: Linear probability models of the start of a team's first qualifying absence on the team's trailing average point differential. Each entry is denominated in percentage points per point of form. Includes team-season and game-number fixed effects. Standard errors, clustered by team-season, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Trailing games	Absence probability per point of form	(SE)	Team-games	Absences
3	+0.026	(0.046)	4,886	268
5	+0.113*	(0.066)	4,332	252
10	-0.089	(0.113)	3,167	169

I test the reversion story by re-running the main specification from Table 5 with the explicit inclusion of controls for pre-event form. The results remain consistent and significant.

Table F.3: The Table 5 point-differential specification re-estimated with pre-event team form added as a moderator. 268 events. Two-step standard errors, clustered by team-season, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_μ	β_σ	τ	β_μ	β_σ
Point differential, pre-window form	-0.974***	-0.170*	+0.590***	-0.974***	-0.324**	+0.356**
added as a moderator	(0.340)	(0.089)	(0.188)	(0.337)	(0.135)	(0.151)

APPENDIX G: ROSTER AND REPLACEMENT QUALITY

Each absence or treatment is moderated by the worker's mean and volatility. However, each absent player is replaced and their minutes are allocated to other players on the roster. Because

mid-season roster changes are heavily constrained, this nearly always means that a player who has not previously played will enter the rotation, minutes will be reallocated to other players who will play more than previously, or some combination of the two. This is a strategic choice made with full information, so it is natural to question how much of the observed effect is a characteristic of the replacements as opposed to the absence itself.

I estimate this by measuring each team's per-team-minute mean and SD for the prior 10 games. I then measure the same value for games within the treatment window and use the difference between the two as a moderator in the main difference-in-differences design.

Table G.1 compares these analyses for point differential (row 2) and win rate (row 4) to the main specification from Table 5 (rows 1 and 3). The differences, for volatility, are negligible, suggesting that the treatment effect is tied to the player who is absent, rather than the players who replace them.

This is to be expected. Teams already play their "best" players (however that is defined) more than others. A replacement is, by definition, more limited in their ability to affect the team's outcomes. Moreover, because teams are all operating with similar roster-building constraints, the quality of replacement players is likely to be very similar. While there is considerable variation in the quality of each team's best player, the 11th best player on every team is likely to be, at best, slightly below average.

Table G.1: Replacement-player controls. Rows (1) and (3) reproduce the Table 5 specifications; rows (2) and (4) add the realized change in the remaining rotation's minutes-weighted mean and volatility as moderators. 268 absence events in every row. Two-step standard errors, clustered by team-season, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_μ	β_σ	τ	β_μ	β_σ
<i>Point differential</i>						
(1) Main specification	-0.974*** (0.337)	-0.169* (0.090)	+0.612*** (0.189)	-0.974*** (0.334)	-0.319** (0.136)	+0.380** (0.154)
(2) Replacement change added as moderators	-0.974*** (0.337)	-0.115 (0.106)	+0.590*** (0.206)	-0.974*** (0.334)	-0.258* (0.141)	+0.376** (0.157)
<i>Win rate</i>						
(3) Main specification	-0.022* (0.012)	-0.0052* (0.0031)	+0.0222*** (0.0072)	-0.022* (0.012)	-0.0088* (0.0046)	+0.0124** (0.0055)
(4) Replacement change added as moderators	-0.022* (0.012)	-0.0023 (0.0036)	+0.0255*** (0.0079)	-0.022* (0.012)	-0.0053 (0.0046)	+0.0131** (0.0058)

APPENDIX H: PLACEBO TESTS AND PERMUTATIONS

As a test to confirm that the observed results are attributable to the absence event, I run two placebo tests. For each, I shift the treatment period to periods when the player was available. The backwards placebo treats the 10 games before the absence as the post-treatment period. The forwards placebo treats the 10 games after the true window as the new post-treatment period. The latter is a less precise test because it only includes absences with fewer than 10 games and which are not impacted by other events. For both placebo tests, the treatment effect (τ) and coefficient on player volatility (β_σ) are insignificant and comparatively small in magnitude as shown in Table H.1. In the case of the more precise backwards placebo, the 95% confidence interval excludes the corresponding main specification estimate.

Table H.1: Placebo windows. Rows (1) and (4) reproduce the Table 5 results (268 events, two-step standard errors clustered by team-season). Rows (2) and (5) are backwards placebos (268 events). Rows (3) and (6) are forwards placebos (116 events). Placebo rows are estimated at the event level with robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_μ	β_σ	τ	β_μ	β_σ
<i>Point differential</i>						
(1) Main specification	-0.974*** (0.337)	-0.169* (0.090)	+0.612*** (0.189)	-0.974*** (0.334)	-0.319** (0.136)	+0.380** (0.154)
(2) Pre-absence window (ten games before)	-0.098 (0.287)	+0.067 (0.077)	+0.077 (0.202)	-0.098 (0.287)	+0.127 (0.112)	+0.012 (0.156)
(3) Post-return window (ten games after the window)	-0.236 (0.552)	+0.112 (0.145)	+0.172 (0.377)	-0.236 (0.550)	+0.330 (0.221)	+0.035 (0.336)
<i>Win rate</i>						
(4) Main specification	-0.022* (0.012)	-0.0052* (0.0031)	+0.0222*** (0.0072)	-0.022* (0.012)	-0.0088* (0.0046)	+0.0124** (0.0055)
(5) Pre-absence window (ten games before)	+0.002 (0.009)	+0.0021 (0.0024)	+0.0038 (0.0068)	+0.002 (0.009)	+0.0060* (0.0035)	-0.0010 (0.0051)
(6) Post-return window (ten games after the window)	-0.006 (0.019)	+0.0016 (0.0050)	+0.0092 (0.0147)	-0.006 (0.019)	+0.0047 (0.0070)	+0.0057 (0.0121)

Further, I use a permutation test to ensure that the observed results are not driven by a handful of players with unique pairs of volatility and mean attributes. Using the same 268 absences, I first randomly assign players (and their mean and volatility) to absences (Permutation 1). I then retain mean and assign a randomly chosen volatility from within the same stratum of mean performance to each event (Permutation 2). This random assignment is performed 2,000 times. Table H.2 shows the frequency with which each permutation produces results with at least as large a magnitude as β_σ .

Table H.2: Permutation tests. Each count is the number, out of 2,000 draws, of permutations which produce $|\beta_\sigma|$ at least as large as observed.

	Observed β_σ	(1) Mean and volatility reshuffled together	(2) Volatility reshuffled within mean quintiles
Point differential, Game Score	+0.612	4	3
Point differential, Box Plus Minus	+0.380	20	21
Win rate, Game Score	+0.0222	5	3
Win rate, Box Plus Minus	+0.0124	53	54

APPENDIX I: TREATMENT WINDOW ROBUSTNESS

When estimating the effect of an absence on team performance, I elected to use absences of at least five games and a treatment window equal to the median absence (11 games). For completeness, I estimate the treatment and moderation effects on alternate choices for both in Table I.1. Rows (1) and (6) reproduce the Table 5 results (268 events). Rows (2) and (7) count as treated only the games the player misses within the 11-game window. Rows (3) and (8) count every game of the absence instead of some fixed treatment window length (268 events each). Rows (4) and (9) raise the qualifying-absence minimum to seven games (264 events). Rows (5) and (10) raise it to ten games (239 events).

When the treatment window is tightened to include only games missed within the 11-game treatment window, the coefficient on Volatility increases in magnitude. The only alternate specification that is materially different from the main results in the manuscript occurs when the treatment window is exactly the games missed, regardless of absence length. Though the sign is consistent for Game Score, the magnitude is smaller. For Box Plus Minus, the sign reverses.

These results are consistent with adaptation on the part of the teams who have lost players. Teams whose players are absent for an extended period of time adjust. Players take on new roles. Playbooks are modified. Strategies and expectations are revised. These adaptations are necessitated by the absence, but they are not the same as the effect of losing a player. Rows (3)

and (8) are, effectively, measuring the team's new equilibrium.

The volatility moderation results for Game Score are not sensitive to the minimum absence length. Moving from a minimum of 5 games to either 7 or 10 produces a coefficient with similar magnitude and significance. When instead using Box Plus Minus, the longer minimum absence produces results that are not statistically significant for Win Rate. In the main specification, this is the least-significant cell, so the attenuation is not unexpected.

Table I.1: Alternative specifications. Two-step standard errors, clustered by team-season, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_{μ}	β_{σ}	τ	β_{μ}	β_{σ}
<i>Point differential</i>						
(1) Main specification	-0.974*** (0.337)	-0.169* (0.090)	+0.612*** (0.189)	-0.974*** (0.334)	-0.319** (0.136)	+0.380** (0.154)
(2) Only games missed, within the window	-0.780** (0.366)	-0.240** (0.105)	+0.708*** (0.207)	-0.780** (0.367)	-0.332** (0.154)	+0.389** (0.172)
(3) All games missed (no window)	-0.132 (0.419)	-0.336*** (0.101)	+0.331 (0.218)	-0.132 (0.403)	-0.573*** (0.129)	-0.008 (0.170)
(4) Absence is 7+ games	-0.934*** (0.337)	-0.177* (0.104)	+0.519** (0.208)	-0.934*** (0.334)	-0.348** (0.155)	+0.334** (0.160)
(5) Absence is 10+ games	-0.651* (0.339)	-0.182* (0.093)	+0.668*** (0.197)	-0.651* (0.340)	-0.339** (0.152)	+0.321** (0.159)
<i>Win rate</i>						
(6) Main specification	-0.022* (0.012)	-0.0052* (0.0031)	+0.0222*** (0.0072)	-0.022* (0.012)	-0.0088* (0.0046)	+0.0124** (0.0055)
(7) Only games missed, within the window	-0.021 (0.013)	-0.0053 (0.0036)	+0.0225*** (0.0081)	-0.021 (0.013)	-0.0070 (0.0051)	+0.0105* (0.0062)
(8) All games missed (no window)	-0.003 (0.015)	-0.0083** (0.0033)	+0.0140 (0.0086)	-0.003 (0.015)	-0.0132*** (0.0045)	-0.0014 (0.0063)
(9) Absence is 7+ games	-0.029** (0.012)	-0.0048 (0.0035)	+0.0158** (0.0076)	-0.029** (0.012)	-0.0092* (0.0051)	+0.0069 (0.0055)
(10) Absence is 10+ games	-0.017 (0.012)	-0.0054* (0.0031)	+0.0194*** (0.0072)	-0.017 (0.012)	-0.0086* (0.0049)	+0.0070 (0.0054)

I.1 Concurrent Absences

Each treatment is the team's first material absence of the season, which means that, by construction, none of the events studied can be caused by other, previous absences. However, I do not exclude absences which are followed by another absence within the treatment window. That is, a second key contributor is injured, traded, or suspended within 11 games of the first absence. Of the 268 qualifying events used in the main specification, estimates for 138 are contaminated by a subsequent absence within the treatment window.

In Table I.2 below, I re-estimate the main two-stage difference-in-differences estimation only on the 130 windows in which there is no immediate follow-on absence. The coefficients retain the same signs, but lose significance except in the case of volatility's effect as measured in Game Score. This is partially an artifact of reduced statistical power, but the "clean" sample is a case of post-treatment selection and is no less biased than the larger, 268 event sample.

Table I.2: The Table 5 point-differential specification re-estimated on the 130 windows in which no second qualifying absence begins. Two-step standard errors, clustered by team-season, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_μ	β_σ	τ	β_μ	β_σ
Point differential, excluding windows	-0.383	-0.139	+0.457*	-0.383	-0.277	+0.233
with a second qualifying absence	(0.431)	(0.125)	(0.237)	(0.428)	(0.170)	(0.184)

APPENDIX J: ENDOGENOUS ABSENCES

A material fraction of qualifying absences (37 of 268) are not injuries, but cases where a player is removed from the roster and, usually, traded to another team. Unlike injuries, which are quasi-random, trades are strategic moves. I have left these events in the main specification for two reasons. First, the timing of a trade requires both the focal organization and one or more counterparties to simultaneously agree to the transaction and the value of the included player(s).

The NBA's salary-matching rules further ensure that the wages of the traded players are similar. An organization cannot unilaterally make a trade and so the timing is still, to some extent, outside their control. Second, for purposes of team-building any strategic behavior would be expected to improve the team's position. This would bias the treatment effect to zero while still leaving useful variation with which to estimate the moderating effect of volatility.

Still, to ensure that endogenous roster choices are not driving observed results, I re-estimate the manuscript's results in Table 5 with these 37 events omitted. Trades do, in expectation, improve the transacting organization. When I remove trades, the average cost of an absence increases and the moderating effect of volatility is substantially unchanged.

Table J.1: Departures. Rows (2) and (4) drop the 37 events in which the absent player appears for another franchise later in the same season, leaving 231 events. Rows (1) and (3) reproduce the Table 5 results. Two-step standard errors, clustered by team-season, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_{μ}	β_{σ}	τ	β_{μ}	β_{σ}
<i>Point differential</i>						
(1) Main specification	-0.974*** (0.337)	-0.169* (0.090)	+0.612*** (0.189)	-0.974*** (0.334)	-0.319** (0.136)	+0.380** (0.154)
(2) Excluding departures	-1.224*** (0.335)	-0.256*** (0.080)	+0.637*** (0.205)	-1.224*** (0.330)	-0.422*** (0.117)	+0.375** (0.165)
<i>Win rate</i>						
(3) Main specification	-0.022* (0.012)	-0.0052* (0.0031)	+0.0222*** (0.0072)	-0.022* (0.012)	-0.0088* (0.0046)	+0.0124** (0.0055)
(4) Excluding departures	-0.031*** (0.012)	-0.0078*** (0.0029)	+0.0225*** (0.0079)	-0.031*** (0.012)	-0.0118*** (0.0041)	+0.0114* (0.0058)